

Ultimate Nutrition

Zinc

Zinc is an essential nutrient required by humans and animals alike. Zinc is a mineral that is vital to healthy living. Zinc is probably most widely known for its ability to prevent colds and shorten the duration of colds. This is because of zinc's powerful ability to strengthen the immune system and improve white blood cell count. It plays a variety of biological roles in the body including catalytic, structural, and regulatory roles. It is required to heal injuries, to maintain fertility in adults and growth in children, synthesize protein, maintain health vision, support immune function, and protect the body against the oxidative effects of free radicals.

Zinc is necessary for the function of many enzymes in the body. Zinc aids in regulating hormones, and it has been shown to help maintain prostate health. Zinc can help prevent and heal cold sores. Zinc aids in the body's absorption of minerals, which means it can help prevent osteoporosis by helping the body absorb calcium. Zinc can improve the health of your skin and hair. The anti-inflammatory benefits of zinc can also help reduce the appearance of acne. Zinc has also been shown to help heal ulcers, and it can help you maintain a healthy digestive tract.

Zinc is essential to keeping the immune system strong. For athletes, this is especially important, since being sidelined with a viral infection, or other illness, can force one to miss valuable workout time. In order to repair micro tears that can occur as a result of strenuous exercise, serious athletes usually require higher amounts of most vitamins and minerals—including zinc. Even a small zinc deficiency can cause health problems, including decreased immunity to diseases such as the common cold.

Exercise increases zinc losses from the human body, and severe zinc deficiency can compromise muscle function. Athletes who do not eat an optimal diet, especially those who are trying to control their weight or use fad diets while exercising strenuously, may become deficient in zinc to the extent that performance or health is compromised.

A double-blind study of male athletes with low blood levels of zinc found that 20 mg per day of zinc improved the flexibility of the red blood cells during exercise, which could benefit blood flow to the muscles.

People who may benefit from a zinc supplement include those who do not consume enough calories, athletes, vegetarians, the elderly, pregnant and lactating women, and people who suffer from digestive diseases. Anyone with a low caloric intake is at higher risk for having a low zinc intake and for developing a zinc deficiency. Therefore, to decrease their risk for developing a zinc deficiency, it is important for individuals in these groups to include sources of zinc including zinc supplements in their daily diet.

SELECTED REFERENCES

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FREQUENTLY ASKED QUESTIONS

1. What is Zinc? What does it do?

Zinc is not a vitamin. It is an essential trace mineral found in almost every cell in the body. It stimulates the activity of over 100 enzymes that help in the biochemical reactions in the body. It supports a healthy immune system and is needed for wound healing. It also helps to maintain taste and smell, and is needed for DNA synthesis.

2. What are the natural sources of zinc?

The natural sources of zinc are red meat, poultry, beans, nuts, whole grain, dairy products, and certain seafood like oysters.

3. Are there any side-effects from Zinc supplements?

Zinc may cause irritation of the stomach, and is best taken with food in order to avoid nausea. Oral zinc should not be taken with foods that will reduce its absorption, such as coffee, bran, or protein. Drinking coffee at the same time as taking zinc can reduce the absorption by as much as half. Even moderate amounts of alcohol impair zinc metabolism and increase its excretion.